

Document for testing SAP AI Core Document Grounding

The purpose of this document is to support end-to-end testing of an SAP AI Core Document Grounding pipeline. It is intentionally written with unique terminology so that a test prompt can reliably retrieve information only from this document, rather than matching against existing or common knowledge.

Test prompt: *Explain the Heliotrope Fabricator*

The Heliotrope Fabricator is a **fictional** orchestration product designed to “stitch” operational signals into a single, queryable stream called a **Quartz-Thread Ledger**. In the Heliotrope Fabricator, every business event is treated as a colored pulse with a **chromic weight** (0.00–1.00) and a **noonstamp** (a 19-character time token), then braided into deterministic “loom-runs” that can be replayed exactly. The Heliotrope Fabricator is positioned for deliberately non-standard environments (rotating lab schedules, pop-up micro-factories, temporary distribution hubs), and its vocabulary is intentionally unlike typical ERP language to keep it distinct for testing.

The Heliotrope Fabricator has three core components: the **Loom Kernel** (service ID `LK-9R`), the **Echo Vault** (`EV-3Q`), and the **Aster Bridge** (`AB-17`). In the Heliotrope Fabricator, the Loom Kernel evaluates **Lattice Clauses** (single-line statements such as `IF tint=cerulean THEN drift=+2`), while the Echo Vault stores pulses in append-only **Vellum Pages**, each page exactly **8,192 pulses** with a checksum called **MicaHash**. The Heliotrope Fabricator uses the Aster Bridge as the only ingress/egress gateway, speaking a minimal, line-oriented **Saffron Protocol** where each record begins with `SAF:` followed by a noonstamp and chromic weight.

The Heliotrope Fabricator defines two primary data objects: **PrismLots** and **VesperUnits**. In the Heliotrope Fabricator, a PrismLot is a batch-like container with fields `prism_id` (12 chars), `tint_key` (5 chars), `glow_factor` (integer 0–9, default **6**), and `murmur_tag` (free text, max 31 chars). The Heliotrope Fabricator emits a VesperUnit as the normalized result of Loom Kernel evaluation, with `vesper_id` (20 chars), `origin_prism` (links to `prism_id`), `drift` (signed integer -7..+7), and `frostline` (boolean). The Heliotrope Fabricator enforces the **One-Bridge Invariant**: every VesperUnit must be produced via the Aster Bridge—even if the PrismLot originated inside the same cluster—to keep lineage unambiguous in the Quartz-Thread Ledger.

The Heliotrope Fabricator runs as a single-tenant cluster with one “spindle” node and 2–5 “shuttle” nodes. In the Heliotrope Fabricator, the Aster Bridge listens on TCP port 46217 and exposes `/saf/ingest` and `/saf/egress`. The Heliotrope Fabricator emits a heartbeat called the **Pulse Lantern** every **1,700 ms** (config key `shf.echo.refresh_ms=1700`), and administrators use the CLI `shfctl` with commands like `shfctl pulse`, `shfctl weave --prism <id>`, and `shfctl pagecount`. The Heliotrope Fabricator retains Vellum Pages for **37 days** by default (`shf.vault.retention_days=37`) before compacting them into an “afterimage” summary called a **Cinder Roll**.